

FEDERATED LEARNING VALIDATION TABLES

2-federation-necessity-validation

FEDERATION NECESSITY VALIDATION — GLOBAL METRICS

split	method	acc	prec_macro	rec_macro	f1_macro	logloss	mcc	kappa	auc_roc	pr_auc
VAL	Centralized_Deep	0.9377	0.9443	0.9017	0.9199	0.2850	0.8449	0.8401	0.9410	0.9614
TEST	Centralized_Deep	0.9335	0.9416	0.8966	0.9155	0.2826	0.8370	0.8313	0.9431	0.9611
VAL	Pooled_Merged_NoClientSim	0.9645	0.9599	0.9549	0.9573	0.2920	0.9148	0.9147	0.9647	0.9735
TEST	Pooled_Merged_NoClientSim	0.9649	0.9582	0.9553	0.9567	0.2892	0.9135	0.9135	0.9650	0.9762
VAL	Local_Only_Isolated	0.9341	0.7747	0.7452	0.7552	0.2173	0.5289	0.5231	0.9062	0.9345
TEST	Local_Only_Isolated	0.9344	0.7752	0.7445	0.7548	0.2112	0.5286	0.5223	0.9101	0.9375
VAL	Local_Ensemble_NoFederation	0.9375	0.9094	0.7625	0.7722	0.2067	0.5906	0.5566	0.9041	0.9393
TEST	Local_Ensemble_NoFederation	0.9394	0.8885	0.7600	0.7686	0.2033	0.5763	0.5487	0.9043	0.9403
VAL	Single_Dataset_Only	0.5104	0.7106	0.5123	0.3447	1.4557	0.0649	0.0637	0.4798	0.5287
TEST	Single_Dataset_Only	0.5105	0.7100	0.5112	0.3437	1.4762	0.0591	0.0617	0.4827	0.5282

3-standard-federated-learning-baseline-validation

GLOBAL VALIDATION METRICS

	variant	acc	precision_macro	recall_macro	f1_macro	logloss	mcc	kappa	auc_roc_macro_ovr	pr_auc_macro
0	FedAvg	0.9722	0.9682	0.9387	0.9475	0.0609	0.9040	0.8990	0.9872	0.9573
1	FedProx	0.9854	0.9804	0.9544	0.9650	0.0501	0.9330	0.9301	0.9898	0.9420
2	SCAFFOLD	0.9649	0.8698	0.8664	0.8601	0.0974	0.7401	0.7339	0.9879	0.9392
3	FedNova	0.9847	0.9791	0.9546	0.9648	0.0456	0.9321	0.9297	0.9906	0.9496
4	MOON	0.9658	0.8702	0.8680	0.8623	0.0843	0.7440	0.7379	0.9881	0.9520

GLOBAL TEST METRICS

	variant	acc	precision_macro	recall_macro	f1_macro	logloss	mcc	kappa	auc_roc_macro_ovr	pr_auc_macro
0	FedAvg	0.9716	0.9671	0.9347	0.9444	0.0628	0.8983	0.8927	0.9863	0.9539
1	FedProx	0.9847	0.9794	0.9502	0.9618	0.0514	0.9274	0.9239	0.9893	0.9397
2	SCAFFOLD	0.9649	0.8698	0.8662	0.8598	0.0983	0.7390	0.7334	0.9874	0.9367
3	FedNova	0.9844	0.9785	0.9508	0.9620	0.0458	0.9272	0.9242	0.9902	0.9460
4	MOON	0.9658	0.8703	0.8679	0.8625	0.0855	0.7443	0.7381	0.9875	0.9487

4-personalized-federated-learning-validation

PERSONALIZED FEDERATED LEARNING VALIDATION — GLOBAL METRICS

split	method	acc	prec_macro	rec_macro	f1_macro	logloss	mcc	kappa	auc_roc	pr_auc
VAL	FedPer	0.8369	0.8524	0.5173	0.4875	1.3842	0.1177	0.0516	0.7009	0.8204
TEST	FedPer	0.8345	0.6623	0.5177	0.4880	1.4021	0.1090	0.0543	0.6973	0.8077
VAL	FedRep	0.8256	0.6875	0.5035	0.4524	1.4707	0.0358	0.0102	0.7126	0.8458
TEST	FedRep	0.8240	0.5090	0.5036	0.4520	1.4885	0.0277	0.0108	0.7103	0.8359
VAL	pFedMe	0.4860	0.4243	0.5240	0.3244	1.9866	0.1093	0.0735	0.7218	0.8044
TEST	pFedMe	0.4851	0.4922	0.5256	0.3260	2.0101	0.1127	0.0766	0.7171	0.7938
VAL	FedBN	0.9289	0.9246	0.8188	0.8413	0.2588	0.7255	0.6935	0.9561	0.9644
TEST	FedBN	0.9276	0.9204	0.8165	0.8376	0.2715	0.7187	0.6859	0.9546	0.9588
VAL	APFL	0.4996	0.5859	0.6687	0.4330	6.6521	0.2805	0.2738	0.6785	0.8082
TEST	APFL	0.4978	0.5780	0.6634	0.4283	6.6398	0.2678	0.2626	0.6838	0.7978

5-feature-selection-necessity-validation

GLOBAL VALIDATION METRICS

	variant	acc	precision_macro	recall_macro	f1_macro	logloss	mcc	kappa	auc_roc_macro_ovr	pr_auc_macro
0	Same Pipeline Without Feature Selection	0.9847	0.9803	0.9524	0.9635	0.0502	0.9306	0.9273	0.9960	0.9773
1	All-Feature Pipeline	0.9705	0.8640	0.8726	0.8675	0.1055	0.7465	0.7457	0.9865	0.9475
2	Random Feature Subset Pipeline	0.9829	0.9796	0.9493	0.9604	0.0545	0.9266	0.9219	0.9956	0.9685
3	Variance-Threshold-Only Filtering	0.9860	0.9812	0.9549	0.9655	0.0537	0.9343	0.9313	0.9966	0.9781
4	Correlation-Pruning-Only Filtering	0.9739	0.8731	0.8805	0.8765	0.0782	0.7628	0.7628	0.9958	0.9651

GLOBAL TEST METRICS

	variant	acc	precision_macro	recall_macro	f1_macro	logloss	mcc	kappa	auc_roc_macro_ovr	pr_auc_macro
0	Same Pipeline Without Feature Selection	0.9840	0.9796	0.9485	0.9606	0.0513	0.9256	0.9215	0.9960	0.9762
1	All-Feature Pipeline	0.9708	0.8644	0.8726	0.8677	0.1073	0.7470	0.7460	0.9867	0.9453
2	Random Feature Subset Pipeline	0.9821	0.9786	0.9454	0.9573	0.0563	0.9212	0.9157	0.9952	0.9659
3	Variance-Threshold-Only Filtering	0.9853	0.9807	0.9512	0.9628	0.0548	0.9296	0.9258	0.9965	0.9775
4	Correlation-Pruning-Only Filtering	0.9739	0.8732	0.8805	0.8765	0.0795	0.7631	0.7628	0.9958	0.9625

6-feature-ranking-mechanism-validation

FEATURE-RANKING VALIDATION — SELECTED FEATURE COUNTS

method	I23Sub	K99Sub	NTD1	NTD2	WII21
MI_Only_Ranking	26	23	15	29	29
GraphRefined_Local	26	23	15	29	29
GraphRefined_NoEdges	26	23	15	29	29
ReliefF_Ranking	26	23	15	29	29
ChiSquare_ANOVA_Univariate	26	23	15	29	29

FEATURE-RANKING MECHANISM VALIDATION — GLOBAL METRICS

split	method	acc	prec_macro	rec_macro	f1_macro	logloss	mcc	kappa	auc_roc	pr_auc
VAL	MI_Only_Ranking	0.9221	0.9380	0.8722	0.8977	0.2762	0.8075	0.7965	0.9355	0.9545
TEST	MI_Only_Ranking	0.9219	0.9370	0.8748	0.8990	0.2730	0.8094	0.7991	0.9366	0.9571
VAL	GraphRefined_Local	0.9415	0.9456	0.9102	0.9257	0.3046	0.8550	0.8516	0.9345	0.9563
TEST	GraphRefined_Local	0.9396	0.9428	0.9100	0.9244	0.3134	0.8521	0.8490	0.9348	0.9588
VAL	GraphRefined_NoEdges	0.9241	0.9380	0.8764	0.9007	0.2563	0.8121	0.8024	0.9380	0.9554
TEST	GraphRefined_NoEdges	0.9231	0.9357	0.8783	0.9011	0.2590	0.8120	0.8030	0.9377	0.9567
VAL	ReliefF_Ranking	0.9319	0.9490	0.8862	0.9110	0.2451	0.8328	0.8229	0.9358	0.9544
TEST	ReliefF_Ranking	0.9305	0.9465	0.8869	0.9105	0.2452	0.8313	0.8219	0.9366	0.9564
VAL	ChiSquare_ANOVA_Univariate	0.9285	0.9476	0.8798	0.9061	0.2671	0.8246	0.8132	0.9281	0.9455
TEST	ChiSquare_ANOVA_Univariate	0.9275	0.9450	0.8815	0.9063	0.2672	0.8240	0.8135	0.9276	0.9453

7-client-selection-and-partitioning-validation

CLIENT-SELECTION AND PARTITIONING VALIDATION — DATASET PROFILE

dataset	train_size	entropy_ratio	default_eacs_clients
WII21	836,124	0.3765	46
NTD1	257,611	0.4196	10
NTD2	39,078	0.9975	12
K99Sub	17,634	0.9967	4
I23Sub	16,201	0.4129	2

CLIENT-SELECTION AND PARTITIONING VALIDATION — SCENARIO SUMMARY

scenario	subset	alpha	n_datasets	avg_clients	datasets
FixedSmall	all	0.700	5	4.000	I23Sub, K99Sub, NTD1, NTD2, WII21
FixedMedium	all	0.700	5	8.000	I23Sub, K99Sub, NTD1, NTD2, WII21
FixedLarge	all	0.700	5	16.000	I23Sub, K99Sub, NTD1, NTD2, WII21
EACS_LowAlpha	all	0.300	5	14.800	I23Sub, K99Sub, NTD1, NTD2, WII21
EACS_HighAlpha	all	1.200	5	14.800	I23Sub, K99Sub, NTD1, NTD2, WII21

CLIENT-SELECTION AND PARTITIONING VALIDATION — GLOBAL METRICS

split	scenario	method	subset	alpha	acc	prec_macro	rec_macro	f1_macro	logloss	mcc	kappa	auc_roc	pr_auc
VAL	FixedSmall	GRIP_DFFI_NO_FS	all	0.7000	0.9589	0.8573	0.8677	0.8594	0.0857	0.7337	0.7316	0.9653	0.8541
TEST	FixedSmall	GRIP_DFFI_NO_FS	all	0.7000	0.9591	0.8573	0.8677	0.8597	0.0854	0.7338	0.7318	0.9663	0.8585
VAL	FixedSmall	FedAvg	all	0.7000	0.8024	0.7328	0.6879	0.6280	1.5607	0.4116	0.3627	0.6216	0.7628
TEST	FixedSmall	FedAvg	all	0.7000	0.7999	0.7304	0.6855	0.6255	1.5760	0.4077	0.3588	0.6218	0.7550
VAL	FixedSmall	FedPer	all	0.7000	0.8742	0.8792	0.7547	0.2820	0.5694	0.5390	0.7920	0.8634	
TEST	FixedSmall	FedPer	all	0.7000	0.8714	0.8737	0.7216	0.7513	0.2858	0.5634	0.5336	0.7875	0.8552
VAL	FixedMedium	GRIP_DFFI_NO_FS	all	0.7000	0.9630	0.8669	0.8702	0.8619	0.0822	0.7430	0.7400	0.9914	0.9660
TEST	FixedMedium	GRIP_DFFI_NO_FS	all	0.7000	0.9632	0.8675	0.8703	0.8621	0.0846	0.7435	0.7404	0.9914	0.9648
VAL	FixedMedium	FedAvg	all	0.7000	0.2300	0.2816	0.4233	0.2035	3.1775	-0.0745	0.0081	0.5116	0.7271
TEST	FixedMedium	FedAvg	all	0.7000	0.2318	0.2831	0.4278	0.2046	3.1831	-0.0702	0.0067	0.5125	0.7170
VAL	FixedMedium	FedPer	all	0.7000	0.8474	0.8565	0.5888	0.5929	1.0663	0.2731	0.2285	0.7108	0.8281
TEST	FixedMedium	FedPer	all	0.7000	0.8460	0.8523	0.5915	0.5937	1.0929	0.2761	0.2318	0.7096	0.8194
VAL	FixedLarge	GRIP_DFFI_NO_FS	all	0.7000	0.9596	0.8625	0.8678	0.8605	0.0657	0.7365	0.7361	0.9833	0.9528
TEST	FixedLarge	GRIP_DFFI_NO_FS	all	0.7000	0.9593	0.8621	0.8676	0.8602	0.0656	0.7362	0.7355	0.9833	0.9522
VAL	FixedLarge	FedAvg	all	0.7000	0.6346	0.5022	0.5857	0.4142	1.0398	0.0946	0.0449	0.6861	0.7518
TEST	FixedLarge	FedAvg	all	0.7000	0.6328	0.4961	0.5845	0.4133	1.0414	0.0933	0.0443	0.6853	0.7458
VAL	FixedLarge	FedPer	all	0.7000	0.8698	0.8865	0.5577	0.5501	0.9362	0.2243	0.1485	0.7644	0.8747
TEST	FixedLarge	FedPer	all	0.7000	0.8682	0.8827	0.5610	0.5544	0.9632	0.2277	0.1583	0.7638	0.8658
VAL	EACS_LowAlpha	GRIP_DFFI_NO_FS	all	0.3000	0.9686	0.8636	0.8710	0.8654	0.0667	0.7441	0.7422	0.9953	0.9709
TEST	EACS_LowAlpha	GRIP_DFFI_NO_FS	all	0.3000	0.9685	0.8705	0.8710	0.8655	0.0675	0.7459	0.7424	0.9952	0.9704
VAL	EACS_LowAlpha	FedAvg	all	0.3000	0.4765	0.4180	0.5001	0.2776	2.9884	0.0093	0.0005	0.6372	0.7468
TEST	EACS_LowAlpha	FedAvg	all	0.3000	0.4749	0.2375	0.5000	0.2769	3.0029	0.0000	0.0000	0.6318	0.7324
VAL	EACS_LowAlpha	FedPer	all	0.3000	0.8305	0.7747	0.5116	0.4693	1.4760	0.0966	0.0406	0.6314	0.7395
TEST	EACS_LowAlpha	FedPer	all	0.3000	0.8292	0.5885	0.5130	0.4714	1.4954	0.0913	0.0449	0.6308	0.7302
VAL	EACS_HighAlpha	GRIP_DFFI_NO_FS	all	1.2000	0.9666	0.8704	0.8721	0.8670	0.0542	0.7509	0.7464	0.9953	0.9654
TEST	EACS_HighAlpha	GRIP_DFFI_NO_FS	all	1.2000	0.9666	0.8706	0.8719	0.8667	0.0548	0.7508	0.7460	0.9955	0.9629
VAL	EACS_HighAlpha	FedAvg	all	1.2000	0.4767	0.5978	0.5002	0.2779	2.5420	0.0118	0.0005	0.6479	0.7394
TEST	EACS_HighAlpha	FedAvg	all	1.2000	0.4751	0.4023	0.5000	0.2771	2.5525	-0.0000	-0.0000	0.6475	0.7366
VAL	EACS_HighAlpha	FedPer	all	1.2000	0.8357	0.8514	0.5457	0.5223	1.4097	0.1935	0.1375	0.6922	0.8133
TEST	EACS_HighAlpha	FedPer	all	1.2000	0.8345	0.8502	0.5480	0.5248	1.4274	0.1968	0.1431	0.6911	0.8045

8-high-heterogeneity-client-validation

GLOBAL VALIDATION METRICS

	variant	acc	precision_macro	recall_macro	f1_macro	logloss	mcc	kappa	auc_roc_macro_ovr	pr_auc_macro
0	High Class-Entropy Datasets	0.9493	0.9320	0.8386	0.8765	0.1840	0.7641	0.7536	0.9755	0.8718
1	Large Datasets Only	0.9771	0.9769	0.9442	0.9516	0.0628	0.9168	0.9110	0.9908	0.9589
2	Extreme Non-IID Dirichlet Split	0.9162	0.4769	0.5180	0.4956	0.7656	0.0335	0.0329	0.9667	0.9195
3	Moderate Non-IID Dirichlet Split	0.9833	0.9796	0.9518	0.9628	0.0579	0.9296	0.9260	0.9903	0.9543
4	Near-IID Client Split (Control)	0.9845	0.9806	0.9531	0.9638	0.0467	0.9318	0.9280	0.9954	0.9681

GLOBAL TEST METRICS

	variant	acc	precision_macro	recall_macro	f1_macro	logloss	mcc	kappa	auc_roc_macro_ovr	pr_auc_macro
0	High Class-Entropy Datasets	0.9473	0.9302	0.8252	0.8665	0.1882	0.7470	0.7338	0.9742	0.8628
1	Large Datasets Only	0.9765	0.9768	0.9405	0.9488	0.0651	0.9122	0.9055	0.9909	0.9574
2	Extreme Non-IID Dirichlet Split	0.9163	0.4770	0.5178	0.4956	0.7708	0.0336	0.0329	0.9627	0.9126
3	Moderate Non-IID Dirichlet Split	0.9830	0.9793	0.9483	0.9602	0.0589	0.9252	0.9209	0.9899	0.9516
4	Near-IID Client Split (Control)	0.9837	0.9796	0.9488	0.9605	0.0478	0.9261	0.9215	0.9950	0.9654

9-backbone-and-core-architecture-validation

BACKBONE AND CORE ARCHITECTURE VALIDATION — GLOBAL METRICS

split	method	acc	prec_macro	rec_macro	f1_macro	logloss	mcc	kappa	auc_roc	pr_auc
VAL	Plain_MLP_Backbone	0.8501	0.8380	0.6008	0.5920	1.4484	0.2996	0.2611	0.6867	0.8168
TEST	Plain_MLP_Backbone	0.8492	0.6566	0.6034	0.5942	1.4638	0.2940	0.2659	0.6863	0.8079
VAL	Plain_Transformer_Backbone	0.8579	0.6087	0.6798	0.6351	0.6560	0.3596	0.3596	0.9102	0.8561
TEST	Plain_Transformer_Backbone	0.8580	0.6088	0.6798	0.6351	0.6557	0.3596	0.3596	0.9103	0.8458
VAL	Shallow_Tabular_Encoder	0.6484	0.7070	0.5854	0.5429	1.5089	0.2821	0.2746	0.7491	0.8741
TEST	Shallow_Tabular_Encoder	0.6499	0.7058	0.5882	0.5455	1.5322	0.2843	0.2771	0.7501	0.8640
VAL	Fully_Shared_Model	0.4765	0.4180	0.5001	0.2776	2.5735	0.0093	0.0005	0.7442	0.7841
TEST	Fully_Shared_Model	0.4749	0.2375	0.5000	0.2769	2.6054	0.0000	0.0000	0.7375	0.7766
VAL	Fully_Private_Model	0.8909	0.9262	0.7227	0.6993	0.2394	0.4785	0.4534	0.8668	0.9513
TEST	Fully_Private_Model	0.8880	0.7421	0.7199	0.6946	0.2482	0.4620	0.4463	0.8584	0.9441

10-diffusion-regularized-shared-backbone-validati

GLOBAL VALIDATION METRICS

	variant	acc	precision_macro	recall_macro	f1_macro	logloss	mcc	kappa	auc_roc_macro_ovr	pr_auc_macro
0	No-Diffusion Backbone	0.9858	0.9797	0.9559	0.9656	0.0490	0.9341	0.9315	0.9928	0.9581
1	Low Diffusion Loss Weight	0.9835	0.9796	0.9527	0.9632	0.0532	0.9304	0.9269	0.9886	0.9485
2	Medium Diffusion Loss Weight	0.9833	0.9796	0.9518	0.9628	0.0579	0.9296	0.9260	0.9903	0.9543
3	High Diffusion Loss Weight	0.9748	0.9694	0.9374	0.9453	0.0619	0.9039	0.8966	0.9870	0.9540
4	Fewer Diffusion Steps	0.9836	0.9789	0.9515	0.9625	0.0505	0.9287	0.9254	0.9941	0.9660
5	More Diffusion Steps	0.9837	0.9809	0.9493	0.9610	0.0566	0.9277	0.9227	0.9871	0.9611
6	High-Heterogeneity Diffusion Setting	0.9162	0.4769	0.5180	0.4956	0.7656	0.0335	0.0329	0.9667	0.9195
7	Noisy Diffusion Setting	0.9828	0.9793	0.9512	0.9622	0.0606	0.9287	0.9249	0.9892	0.9576

GLOBAL TEST METRICS

	variant	acc	precision_macro	recall_macro	f1_macro	logloss	mcc	kappa	auc_roc_macro_ovr	pr_auc_macro
0	No-Diffusion Backbone	0.9852	0.9789	0.9521	0.9628	0.0504	0.9291	0.9258	0.9925	0.9552
1	Low Diffusion Loss Weight	0.9830	0.9792	0.9488	0.9604	0.0548	0.9257	0.9213	0.9882	0.9455
2	Medium Diffusion Loss Weight	0.9830	0.9793	0.9483	0.9602	0.0589	0.9252	0.9209	0.9899	0.9516
3	High Diffusion Loss Weight	0.9743	0.9758	0.9338	0.9427	0.0630	0.9001	0.8914	0.9867	0.9515
4	Fewer Diffusion Steps	0.9829	0.9785	0.9475	0.9595	0.0517	0.9236	0.9195	0.9939	0.9634
5	More Diffusion Steps	0.9829	0.9801	0.9455	0.9580	0.0584	0.9225	0.9168	0.9865	0.9581
6	High-Heterogeneity Diffusion Setting	0.9163	0.4770	0.5178	0.4956	0.7708	0.0336	0.0329	0.9627	0.9126
7	Noisy Diffusion Setting	0.9826	0.9790	0.9477	0.9596	0.0619	0.9244	0.9199	0.9887	0.9547

11-routing-and-personalization-validation

ROUTING AND PERSONALIZATION VALIDATION — GLOBAL METRICS

split	method	acc	prec_macro	rec_macro	f1_macro	logloss	mcc	kappa	auc_roc	pr_auc
VAL	No_Routing_Setting	0.8850	0.9080	0.7167	0.6879	1.3285	0.4663	0.4426	0.8316	0.8857
TEST	No_Routing_Setting	0.8826	0.8960	0.7148	0.6843	1.3527	0.4560	0.4363	0.8268	0.8776
VAL	Single_Tokenizer_Setting	0.1672	0.0836	0.5000	0.1276	5.9046	0.0000	0.0000	0.9328	0.8661
TEST	Single_Tokenizer_Setting	0.1687	0.0844	0.5000	0.1289	5.8810	0.0000	0.0000	0.9310	0.8552
VAL	Layer_Only_Personalization	0.8369	0.8524	0.5173	0.4875	1.3842	0.1177	0.0516	0.7009	0.8204
TEST	Layer_Only_Personalization	0.8345	0.6623	0.5177	0.4880	1.4021	0.1090	0.0543	0.6973	0.8077
VAL	All_Shared_Routing	0.8500	0.7902	0.5869	0.5817	1.3974	0.2655	0.2308	0.6686	0.8079
TEST	All_Shared_Routing	0.8480	0.8022	0.5866	0.5795	1.4187	0.2556	0.2277	0.6667	0.7986
VAL	All_Private_Routing	0.8907	0.9114	0.7522	0.7469	0.2736	0.5993	0.5486	0.9175	0.9560
TEST	All_Private_Routing	0.8883	0.9050	0.7516	0.7448	0.2881	0.5942	0.5452	0.9091	0.9481

12-classical-machine-learning-validation

GLOBAL VALIDATION METRICS

variant	acc	precision_macro	recall_macro	f1_macro	logloss	mcc	kappa	auc_roc_macro_ovr	pr_auc_macro
SVM	0.9894	0.9820	0.9869	0.9843	0.0419	0.9689	0.9687	0.9969	0.9842
Logistic Regression	0.9744	0.9462	0.9670	0.9546	0.0801	0.9117	0.9094	0.9907	0.9671
GaussianNB	0.9496	0.9247	0.9311	0.9251	0.5038	0.8542	0.8504	0.9701	0.8798
AdaBoost	0.9896	0.9879	0.9887	0.9883	0.4003	0.9766	0.9765	0.9981	0.9967
LDA	0.9766	0.9699	0.9451	0.9555	0.1821	0.9133	0.9111	0.9830	0.9441
Ridge Classifier	0.9604	0.9154	0.9522	0.9273	0.3663	0.8631	0.8559	0.9857	0.9397

GLOBAL TEST METRICS

variant	acc	precision_macro	recall_macro	f1_macro	logloss	mcc	kappa	auc_roc_macro_ovr	pr_auc_macro
SVM	0.9897	0.9829	0.9887	0.9856	0.0397	0.9714	0.9712	0.9968	0.9868
Logistic Regression	0.9723	0.9437	0.9650	0.9521	0.0820	0.9070	0.9045	0.9905	0.9665
GaussianNB	0.9502	0.9264	0.9311	0.9265	0.5158	0.8562	0.8531	0.9683	0.8826
AdaBoost	0.9888	0.9866	0.9881	0.9873	0.3995	0.9748	0.9747	0.9984	0.9969
LDA	0.9759	0.9691	0.9453	0.9556	0.1884	0.9130	0.9112	0.9822	0.9405
Ridge Classifier	0.9598	0.9165	0.9498	0.9269	0.3661	0.8620	0.8551	0.9855	0.9351

13-federated-round-budget-validation

GLOBAL VALIDATION METRICS

	variant	acc	precision_macro	recall_macro	f1_macro	logloss	mcc	kappa	auc_roc_macro_ovr	pr_auc_macro
0	1 Round	0.9760	0.9727	0.9443	0.9520	0.0665	0.9144	0.9097	0.9914	0.9519
1	2 Rounds	0.9833	0.9796	0.9518	0.9628	0.0579	0.9296	0.9260	0.9903	0.9543
2	5 Rounds	0.9833	0.9796	0.9518	0.9628	0.0579	0.9296	0.9260	0.9903	0.9543
3	10 Rounds	0.9883	0.9837	0.9637	0.9723	0.0369	0.9464	0.9448	0.9941	0.9548
4	Early-Stop Selected Round	0.9833	0.9796	0.9518	0.9628	0.0579	0.9296	0.9260	0.9903	0.9543

GLOBAL TEST METRICS

	variant	acc	precision_macro	recall_macro	f1_macro	logloss	mcc	kappa	auc_roc_macro_ovr	pr_auc_macro
0	1 Round	0.9756	0.9727	0.9408	0.9493	0.0683	0.9102	0.9045	0.9914	0.9495
1	2 Rounds	0.9830	0.9793	0.9483	0.9602	0.0589	0.9252	0.9209	0.9899	0.9516
2	5 Rounds	0.9830	0.9793	0.9483	0.9602	0.0589	0.9252	0.9209	0.9899	0.9516
3	10 Rounds	0.9877	0.9833	0.9605	0.9701	0.0378	0.9425	0.9403	0.9937	0.9532
4	Early-Stop Selected Round	0.9830	0.9793	0.9483	0.9602	0.0589	0.9252	0.9209	0.9899	0.9516

14-leave-one-dataset-out-cross-corpus-generalizat

FINAL REPORT — LEAVE-ONE-DATASET-OUT CROSS-CORPUS GENERALIZATION

fold	held_out_dataset	trained_on	split	acc	prec_macro	rec_macro	f1_macro	logloss	mcc	kappa	auc_roc_macro	pr_auc	prec_weighted	rec_weighted	f1_weighted	auc_roc_micro	auc_roc_weighted	ppv_macro	npv_macro	ppv_weighted	npv_weighted	ppv_positive	npv_negative
fold_1	I23Sub	K99Sub, NTD1, NTD2, WII21	VAL	0.9971	0.9921	0.9890	0.9905	0.0136	0.9810	0.9810	0.9961	0.9849	0.9971	0.9971	0.9971	0.9961	0.9961	0.9921	0.9921	0.9971	0.9870	0.9860	0.9981
fold_1	I23Sub	K99Sub, NTD1, NTD2, WII21	TEST	0.9954	0.9926	0.9770	0.9847	0.0220	0.9695	0.9693	0.9949	0.9762	0.9954	0.9954	0.9954	0.9949	0.9949	0.9926	0.9926	0.9954	0.9898	0.9892	0.9959
fold_2	K99Sub	I23Sub, NTD1, NTD2, WII21	VAL	0.9701	0.9704	0.9696	0.9699	0.1034	0.9399	0.9399	0.9923	0.9927	0.9701	0.9927	0.9701	0.9923	0.9923	0.9704	0.9704	0.9701	0.9706	0.9671	0.9736
fold_2	K99Sub	I23Sub, NTD1, NTD2, WII21	TEST	0.9685	0.9694	0.9675	0.9683	0.1061	0.9369	0.9366	0.9915	0.9920	0.9687	0.9685	0.9685	0.9915	0.9915	0.9694	0.9694	0.9687	0.9700	0.9598	0.9790
fold_3	NTD1	I23Sub, K99Sub, NTD2, WII21	VAL	0.9799	0.9356	0.9354	0.9355	0.0591	0.8710	0.8710	0.9924	0.9144	0.9799	0.9799	0.9799	0.9924	0.9924	0.9356	0.9356	0.9799	0.8913	0.8823	0.9890
fold_3	NTD1	I23Sub, K99Sub, NTD2, WII21	TEST	0.9782	0.9305	0.9293	0.9299	0.0622	0.8598	0.8598	0.9918	0.9084	0.9782	0.9782	0.9782	0.9918	0.9918	0.9305	0.9305	0.9782	0.8828	0.8730	0.9880
fold_4	NTD2	I23Sub, K99Sub, NTD1, WII21	VAL	0.9808	0.9807	0.9807	0.9807	0.0771	0.9614	0.9614	0.9953	0.9938	0.9808	0.9808	0.9808	0.9953	0.9953	0.9807	0.9807	0.9808	0.9806	0.9790	0.9824
fold_4	NTD2	I23Sub, K99Sub, NTD1, WII21	TEST	0.9835	0.9833	0.9837	0.9835	0.0710	0.9670	0.9669	0.9954	0.9937	0.9836	0.9835	0.9835	0.9954	0.9954	0.9833	0.9833	0.9836	0.9830	0.9786	0.9880
fold_5	WII21	I23Sub, K99Sub, NTD1, NTD2	VAL	1.0000	1.0000	1.0000	1.0000	0.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
fold_5	WII21	I23Sub, K99Sub, NTD1, NTD2	TEST	1.0000	1.0000	1.0000	1.0000	0.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
summary	global_weighted	all leave-one-dataset-out folds	VAL	0.9944	0.9846	0.9845	0.9845	0.0174	0.9691	0.9691	0.9980	0.9806	0.9944	0.9944	0.9944	0.9980	0.9980	0.9846	0.9846	0.9944	0.9747	0.9726	0.9966
summary	global_weighted	all leave-one-dataset-out folds	TEST	0.9941	0.9835	0.9830	0.9833	0.0180	0.9666	0.9666	0.9978	0.9791	0.9941	0.9941	0.9941	0.9978	0.9978	0.9835	0.9835	0.9941	0.9730	0.9705	0.9966

15-split-ratio-robustness-validation

FINAL REPORT - SPLIT-RATIO ROBUSTNESS VALIDATION

split_ratio	split	dataset	acc	prec_macro	rec_macro	f1_macro	logloss	mcc	kappa	auc_roc_macro	pr_auc	prec_weighted	rec_weighted	f1_weighted	auc_roc_micro	auc_roc_weighted	ppv_macro	npv_macro	ppv_weighted	npv_weighted	ppv_positive	npv_negative
60_20_20	VAL	I23Sub	0.9974	0.9986	0.9844	0.9914	0.2095	0.9829	0.9827	1.0000	0.9999	0.9974	0.9974	0.9974	1.0000	1.0000	0.9986	0.9986	0.9974	0.9998	1.0000	0.9972
60_20_20	TEST	I23Sub	0.9972	0.9985	0.9831	0.9906	0.2108	0.9815	0.9813	1.0000	1.0000	0.9972	0.9972	0.9972	1.0000	1.0000	0.9985	0.9985	0.9972	0.9997	1.0000	0.9969
60_20_20	VAL	K99Sub	0.8851	0.9104	0.8769	0.8812	0.3402	0.7865	0.7655	0.9776	0.9808	0.9045	0.8827	0.8827	0.9776	0.9776	0.9104	0.9104	0.9045	0.9162	0.9966	0.8241
60_20_20	TEST	K99Sub	0.8873	0.9121	0.8792	0.8836	0.3352	0.7906	0.7700	0.9825	0.9848	0.9063	0.8873	0.8850	0.9825	0.9825	0.9121	0.9121	0.9063	0.9179	0.9978	0.8265
60_20_20	VAL	NTD1	0.9652	0.9416	0.8230	0.8712	0.1536	0.7554	0.7430	0.9757	0.8426	0.9639	0.9652	0.9625	0.9757	0.9757	0.9416	0.9416	0.9639	0.9193	0.9147	0.9685
60_20_20	TEST	NTD1	0.9659	0.9427	0.8267	0.8741	0.1507	0.7605	0.7487	0.9766	0.8468	0.9646	0.9659	0.9633	0.9766	0.9766	0.9427	0.9427	0.9646	0.9207	0.9162	0.9691
60_20_20	VAL	NTD2	0.7689	0.8326	0.7815	0.7622	0.5770	0.6119	0.5483	0.9486	0.9277	0.8420	0.7689	0.7598	0.9486	0.9486	0.8326	0.8326	0.8420	0.8232	0.6717	0.9935
60_20_20	TEST	NTD2	0.7720	0.8340	0.7844	0.7656	0.5765	0.6163	0.5541	0.9492	0.9288	0.8433	0.7720	0.7633	0.9492	0.9492	0.8340	0.8340	0.8433	0.8247	0.6747	0.9933
60_20_20	VAL	WI121	1.0000	1.0000	1.0000	1.0000	0.0001	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
60_20_20	TEST	WI121	1.0000	1.0000	1.0000	1.0000	0.0001	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
60_20_20	VAL	global_weighted	0.9828	0.9801	0.9515	0.9617	0.0613	0.9295	0.9243	0.9926	0.9625	0.9853	0.9828	0.9819	0.9926	0.9926	0.9801	0.9801	0.9853	0.9750	0.9701	0.9901
60_20_20	TEST	global_weighted	0.9831	0.9804	0.9524	0.9625	0.0606	0.9309	0.9258	0.9929	0.9636	0.9855	0.9831	0.9822	0.9929	0.9929	0.9804	0.9804	0.9855	0.9754	0.9706	0.9903
80_10_10	VAL	I23Sub	0.9399	0.9693	0.6380	0.7005	0.1032	0.5090	0.4115	0.9959	0.9605	0.9436	0.9399	0.9238	0.9959	0.9959	0.9693	0.9693	0.9436	0.9949	1.0000	0.9385
80_10_10	TEST	I23Sub	0.9430	0.9707	0.6580	0.7251	0.1079	0.5455	0.4586	0.9931	0.9426	0.9463	0.9430	0.9290	0.9931	0.9931	0.9707	0.9707	0.9463	0.9951	1.0000	0.9414
80_10_10	VAL	K99Sub	0.9154	0.9294	0.9097	0.9136	0.2822	0.8389	0.8283	0.9865	0.9876	0.9252	0.9154	0.9145	0.9865	0.9865	0.9294	0.9294	0.9252	0.9337	0.9918	0.8671
80_10_10	TEST	K99Sub	0.9218	0.9336	0.9167	0.9203	0.2771	0.8501	0.8414	0.9835	0.9856	0.9298	0.9218	0.9211	0.9835	0.9835	0.9336	0.9336	0.9298	0.9374	0.9900	0.8771
80_10_10	VAL	NTD1	0.9150	0.4575	0.5000	0.4778	0.3358	0.0000	0.0000	0.9710	0.8212	0.8372	0.9150	0.8744	0.9710	0.9710	0.4575	0.4575	0.8372	0.0778	0.0000	0.9150
80_10_10	TEST	NTD1	0.9150	0.4575	0.5000	0.4778	0.3315	0.0000	0.0000	0.9731	0.8414	0.8372	0.9150	0.8744	0.9731	0.9731	0.4575	0.4575	0.8372	0.0778	0.0000	0.9150
80_10_10	VAL	NTD2	0.7629	0.8381	0.7486	0.7414	0.5614	0.5798	0.5109	0.8930	0.9015	0.8296	0.7629	0.7458	0.8930	0.8930	0.8381	0.8381	0.8296	0.8466	0.9837	0.6926
80_10_10	TEST	NTD2	0.7693	0.8413	0.7554	0.7495	0.5570	0.5905	0.5246	0.9017	0.9094	0.8329	0.7693	0.7536	0.9017	0.9017	0.8413	0.8413	0.8329	0.8497	0.9841	0.6985
80_10_10	VAL	WI121	0.9999	1.0000	0.9995	0.9998	0.0005	0.9995	0.9995	1.0000	1.0000	0.9999	0.9999	0.9999	1.0000	1.0000	1.0000	1.0000	0.9999	1.0000	1.0000	0.9999
80_10_10	TEST	WI121	0.9999	0.9999	0.9992	0.9996	0.0009	0.9991	0.9991	1.0000	1.0000	0.9999	0.9999	0.9999	1.0000	1.0000	0.9999	0.9999	0.9999	1.0000	1.0000	0.9999
80_10_10	VAL	global_weighted	0.9711	0.8733	0.8744	0.8704	0.0990	0.7555	0.7517	0.9897	0.9564	0.9564	0.9711	0.9613	0.9897	0.9897	0.8733	0.8733	0.9564	0.7901	0.7785	0.9680
80_10_10	TEST	global_weighted	0.9714	0.8734	0.8748	0.8710	0.0982	0.7563	0.7527	0.9904	0.9609	0.9566	0.9714	0.9617	0.9904	0.9904	0.8734	0.8734	0.9566	0.7903	0.7785	0.9684

16-multi-seed-statistical-confidence-validation

FINAL REPORT - MULTI-SEED PER-SEED TEST METRICS

seed	split	dataset	acc	prec_macro	rec_macro	f1_macro	logloss	mcc	kappa	auc_roc_macro	pr_auc	prec_weighted	rec_weighted	f1_weighted	auc_roc_micro	auc_roc_weighted	ppv_macro	npv_macro	ppv_weighted	npv_weighted	ppv_positive	npv_negative
11	TEST	I23Sub	0.9948	0.9955	0.9703	0.9825	0.3316	0.9655	0.9650	0.9737	0.9698	0.9948	0.9948	0.9947	0.9737	0.9737	0.9955	0.9955	0.9948	0.9962	0.9963	0.9947
11	TEST	K99Sub	0.8042	0.8337	0.7936	0.7952	0.4806	0.6260	0.5988	0.9317	0.8802	0.8280	0.8042	0.7981	0.9317	0.9317	0.8337	0.8337	0.8280	0.8393	0.9175	0.7499
11	TEST	NTD1	0.9651	0.9401	0.8235	0.8710	0.2031	0.7546	0.7425	0.9700	0.8376	0.9637	0.9651	0.9625	0.9700	0.9700	0.9401	0.9401	0.9637	0.9164	0.9116	0.9686
11	TEST	NTD2	0.8555	0.8762	0.8626	0.8548	0.5852	0.7387	0.7143	0.9672	0.9555	0.8823	0.8555	0.8542	0.9672	0.9672	0.8762	0.8762	0.8823	0.8701	0.7721	0.9803
11	TEST	WII21	1.0000	1.0000	0.9999	0.9999	0.0002	0.9999	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
11	TEST	global_weighted	0.9844	0.9800	0.9528	0.9633	0.0765	0.9308	0.9269	0.9854	0.9909	0.9854	0.9844	0.9837	0.9800	0.9909	0.9800	0.9800	0.9854	0.9747	0.9715	0.9885
22	TEST	I23Sub	0.9329	0.8413	0.6413	0.6917	0.1366	0.4392	0.3905	0.9554	0.6651	0.9230	0.9329	0.9191	0.9554	0.9554	0.8413	0.8413	0.9230	0.7596	0.7434	0.9393
22	TEST	K99Sub	0.8732	0.8992	0.8648	0.8688	0.3211	0.7632	0.7412	0.9707	0.9574	0.8934	0.8732	0.8704	0.9707	0.9707	0.8992	0.8992	0.8934	0.9051	0.8129	0.9856
22	TEST	NTD1	0.9246	0.9366	0.5592	0.5859	0.2129	0.3215	0.1961	0.9759	0.8644	0.9264	0.9246	0.8967	0.9264	0.9759	0.9366	0.9366	0.9264	0.9469	0.9490	0.9243
22	TEST	NTD2	0.8199	0.8641	0.8095	0.8103	0.5238	0.6715	0.6314	0.9722	0.9618	0.8575	0.8199	0.8128	0.9722	0.9722	0.8641	0.8641	0.8575	0.8708	0.9773	0.7510
22	TEST	WII21	1.0000	1.0000	0.9998	0.9999	0.0003	0.9998	0.9998	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
22	TEST	global_weighted	0.9744	0.9777	0.8891	0.8959	0.0715	0.8277	0.7977	0.9927	0.9635	0.9763	0.9744	0.9678	0.9927	0.9927	0.9777	0.9777	0.9763	0.9792	0.9816	0.9739
33	TEST	I23Sub	0.9905	0.9949	0.9429	0.9671	0.1578	0.9363	0.9343	0.9897	0.9990	0.9906	0.9897	0.9905	0.9902	0.9897	0.9949	0.9949	0.9906	0.9991	0.9897	1.0000
33	TEST	K99Sub	0.8391	0.8711	0.8290	0.8320	0.4240	0.9653	0.6706	0.9653	0.9628	0.8648	0.8391	0.8344	0.9653	0.9653	0.8711	0.8711	0.8648	0.8774	0.7777	0.9646
33	TEST	NTD1	0.9634	0.9282	0.8220	0.8659	0.1670	0.7426	0.7324	0.9346	0.8097	0.9616	0.9634	0.9608	0.9346	0.9346	0.9282	0.9282	0.9616	0.8948	0.8879	0.9684
33	TEST	NTD2	0.8278	0.8448	0.8211	0.8231	0.5578	0.6654	0.6501	0.8099	0.8249	0.8411	0.8278	0.8248	0.8099	0.8099	0.8448	0.8448	0.8411	0.8485	0.9080	0.7815
33	TEST	WII21	1.0000	1.0000	0.9999	1.0000	0.0001	0.9999	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
33	TEST	global_weighted	0.9836	0.9769	0.9513	0.9614	0.0642	0.9265	0.9232	0.9785	0.9515	0.9840	0.9836	0.9828	0.9785	0.9785	0.9769	0.9769	0.9840	0.9698	0.9687	0.9852
44	TEST	I23Sub	0.9554	0.9768	0.7318	0.8049	0.1128	0.6649	0.6132	0.9993	0.9935	0.9574	0.9554	0.9477	0.9993	0.9993	0.9768	0.9768	0.9574	0.9961	1.0000	0.9536
44	TEST	K99Sub	0.8907	0.8919	0.8887	0.8898	0.4060	0.7806	0.7797	0.9650	0.9565	0.8912	0.8907	0.8905	0.9650	0.8919	0.8919	0.8919	0.8912	0.8927	0.8812	0.9027
44	TEST	NTD1	0.9150	0.4575	0.5000	0.4778	0.1468	0.0000	0.0000	0.9860	0.8847	0.8372	0.9150	0.8744	0.9860	0.9860	0.4575	0.4575	0.8372	0.0778	0.0000	0.9150
44	TEST	NTD2	0.5616	0.7224	0.5349	0.4242	0.6013	0.1761	0.0734	0.9528	0.9688	0.7121	0.5616	0.4407	0.9528	0.9528	0.7224	0.7224	0.7121	0.7326	0.5474	0.8974
44	TEST	WII21	0.9995	0.9997	0.9966	0.9982	0.0016	0.9964	0.9964	1.0000	1.0000	0.9995	0.9995	0.9995	1.0000	1.0000	0.9997	0.9997	0.9995	0.9995	1.0000	1.0000
44	TEST	global_weighted	0.9639	0.8688	0.8662	0.8597	0.0614	0.7410	0.7368	0.9948	0.9727	0.9518	0.9639	0.9508	0.9948	0.9948	0.8688	0.8688	0.9518	0.7857	0.7618	0.9757
55	TEST	I23Sub	0.7560	0.6225	0.8512	0.6217	0.2711	0.4149	0.3046	0.9735	0.8787	0.9338	0.7560	0.8097	0.9338	0.9735	0.6225	0.6225	0.9338	0.3112	0.2493	0.9958
55	TEST	K99Sub	0.8788	0.8963	0.8718	0.8755	0.4877	0.7677	0.7533	0.8537	0.7570	0.8917	0.8788	0.8769	0.8537	0.8537	0.8963	0.8963	0.8917	0.9009	0.8282	0.9644
55	TEST	NTD1	0.9150	0.4575	0.5000	0.4778	0.1392	0.0000	0.0000	0.9885	0.8923	0.8372	0.9150	0.8744	0.9885	0.9885	0.4575	0.4575	0.8372	0.0778	0.0000	0.9150
55	TEST	NTD2	0.4709	0.5687	0.5001	0.3205	0.6598	0.0052	0.0002	0.9382	0.9175	0.5745	0.4709	0.3018	0.9382	0.9382	0.5687	0.5687	0.5745	0.4708	0.6667	0.6667
55	TEST	WII21	0.9999	1.0000	0.9996	0.9998	0.0003	0.9996	0.9996	1.0000	1.0000	0.9999	0.9999	0.9999	1.0000	1.0000	1.0000	1.0000	0.9999	1.0000	1.0000	0.9999
55	TEST	global_weighted	0.9582	0.8589	0.8686	0.8546	0.0642	0.7339	0.7320	0.9928	0.9681	0.9472	0.9582	0.9443	0.9928	0.9928	0.8589	0.8589	0.9472	0.7707	0.7484	0.9694

FINAL REPORT - MULTI-SEED STATISTICAL CONFIDENCE SUMMARY

split	dataset	n_s eed s	acc	prec_macro	rec_macro	f1_macro	logloss	mcc	kappa	auc_roc_macro	pr_auc	prec_weighted	rec_weighted	f1_weighted	auc_roc_micro	auc_roc_weighted	ppv_macro	npv_macro	ppv_weighted	npv_weighted	ppv_positive	npv_negative
TEST	I23Sub	5	0.9259 +/- 0.0983	0.8862 +/- 0.1609	0.8275 +/- 0.1397	0.8136 +/- 0.1611	0.2020 +/- 0.0946	0.6842 +/- 0.2625	0.6415 +/- 0.3032	0.9783 +/- 0.0169	0.9012 +/- 0.1406	0.9599 +/- 0.0324	0.9259 +/- 0.0983	0.9323 +/- 0.0753	0.9783 +/- 0.0169	0.9783 +/- 0.0169	0.8862 +/- 0.1609	0.8862 +/- 0.1609	0.9599 +/- 0.0324	0.8125 +/- 0.2985	0.7958 +/- 0.3244	0.9767 +/- 0.0281
TEST	K99Sub	5	0.8572 +/- 0.0353	0.8785 +/- 0.0273	0.8496 +/- 0.0381	0.8523 +/- 0.0384	0.4239 +/- 0.0674	0.7273 +/- 0.0648	0.7087 +/- 0.0735	0.9373 +/- 0.0492	0.9028 +/- 0.0884	0.8738 +/- 0.0282	0.8572 +/- 0.0353	0.8540 +/- 0.0375	0.9373 +/- 0.0492	0.9373 +/- 0.0492	0.8785 +/- 0.0273	0.8785 +/- 0.0273	0.8738 +/- 0.0282	0.8831 +/- 0.0266	0.8435 +/- 0.0557	0.9134 +/- 0.0965
TEST	NTD1	5	0.9366 +/- 0.0255	0.7440 +/- 0.2616	0.6409 +/- 0.1677	0.6557 +/- 0.1992	0.1738 +/- 0.0330	0.3637 +/- 0.3751	0.3342 +/- 0.3767	0.9710 +/- 0.0217	0.8577 +/- 0.0342	0.9052 +/- 0.0638	0.9366 +/- 0.0255	0.9137 +/- 0.0447	0.9710 +/- 0.0217	0.9710 +/- 0.0217	0.7440 +/- 0.2616	0.7440 +/- 0.2616	0.9052 +/- 0.0638	0.5827 +/- 0.4613	0.5497 +/- 0.5023	0.9383 +/- 0.0279
TEST	NTD2	5	0.7071 +/- 0.1777	0.7752 +/- 0.1307	0.7056 +/- 0.1733	0.6466 +/- 0.2535	0.5856 +/- 0.0508	0.4514 +/- 0.3360	0.4139 +/- 0.3466	0.9281 +/- 0.0674	0.9257 +/- 0.0598	0.7735 +/- 0.1293	0.7071 +/- 0.1777	0.6469 +/- 0.2568	0.9281 +/- 0.0674	0.9281 +/- 0.0674	0.7752 +/- 0.1307	0.7752 +/- 0.1307	0.7735 +/- 0.1293	0.7770 +/- 0.1326	0.7351 +/- 0.2208	0.8154 +/- 0.1238
TEST	WII21	5	0.9999 +/- 0.0002	0.9999 +/- 0.0001	0.9992 +/- 0.0014	0.9996 +/- 0.0008	0.0005 +/- 0.0006	0.9991 +/- 0.0015	0.9991 +/- 0.0015	1.0000 +/- 0.0000	1.0000 +/- 0.0000	0.9999 +/- 0.0002	0.9999 +/- 0.0002	0.9999 +/- 0.0002	1.0000 +/- 0.0000	1.0000 +/- 0.0000	0.9999 +/- 0.0001	0.9999 +/- 0.0001	0.9999 +/- 0.0002	1.0000 +/- 0.0000	0.9999 +/- 0.0002	1.0000 +/- 0.0000
TEST	global_weighted	5	0.9729 +/- 0.0117	0.9325 +/- 0.0628	0.9056 +/- 0.0433	0.9070 +/- 0.0530	0.0675 +/- 0.0062	0.8320 +/- 0.0957	0.8233 +/- 0.0964	0.9899 +/- 0.0065	0.9633 +/- 0.0080	0.9689 +/- 0.0181	0.9729 +/- 0.0117	0.9659 +/- 0.0180	0.9899 +/- 0.0065	0.9899 +/- 0.0065	0.9325 +/- 0.0628	0.9325 +/- 0.0628	0.9689 +/- 0.0181	0.8960 +/- 0.1077	0.8864 +/- 0.1200	0.9785 +/- 0.0080

17-input-corruption-and-missingness-robustness

FINAL REPORT - INPUT CORRUPTION AND MISSINGNESS ROBUSTNESS VALIDATION

variant	corruption_type	target_rate	actual_rate	split	dataset	corrupted_cells	acc	prec_macro	rec_macro	f1_macro	logloss	mcc	kappa	auc_roc_macro	pr_auc	prec_weighted	rec_weighted	f1_weighted	auc_roc_micro	auc_roc_weighted	ppv_macro	npv_macro	ppv_weighted	npv_weighted	ppv_positive	npv_negative
numeric_missing_10	numeric_missing	0.1000	0.0000	VAL	I23Sub	0	0.9185	0.9592	0.5087	0.4958	0.1088	0.1263	0.0314	0.9955	0.9831	0.9251	0.9185	0.8809	0.9955	0.9955	0.9592	0.9592	0.9251	0.9932	1.0000	0.9184
numeric_missing_10	numeric_missing	0.1000	0.0000	TEST	I23Sub	0	0.9188	0.9593	0.5121	0.5024	0.1111	0.1492	0.0435	0.9930	0.9730	0.9254	0.9188	0.8818	0.9930	0.9930	0.9593	0.9593	0.9254	0.9932	1.0000	0.9186
numeric_missing_10	numeric_missing	0.1000	0.1000	VAL	K99Sub	6046	0.9198	0.9276	0.9156	0.9186	0.3090	0.8432	0.8376	0.9781	0.9763	0.9247	0.9198	0.9781	0.9763	0.9930	0.9593	0.9276	0.9306	0.8844	0.9709	
numeric_missing_10	numeric_missing	0.1000	0.0975	TEST	K99Sub	5896	0.9217	0.9306	0.9172	0.9204	0.3102	0.8477	0.8413	0.9780	0.9741	0.9274	0.9198	0.9780	0.9741	0.9930	0.9593	0.9274	0.9338	0.8833	0.9778	
numeric_missing_10	numeric_missing	0.1000	0.1000	VAL	NTD1	38660	0.9526	0.9331	0.7430	0.8062	0.1648	0.6488	0.6148	0.9778	0.7996	0.9510	0.9526	0.9460	0.9778	0.9778	0.9331	0.9331	0.9274	0.9153	0.9116	0.9546
numeric_missing_10	numeric_missing	0.1000	0.0992	TEST	NTD1	38322	0.9515	0.9341	0.7345	0.7989	0.1714	0.6381	0.6005	0.9768	0.7947	0.9499	0.9515	0.9443	0.9768	0.9768	0.9341	0.9341	0.9499	0.9183	0.9151	0.9531
numeric_missing_10	numeric_missing	0.1000	0.0987	VAL	NTD2	6612	0.7940	0.8460	0.7823	0.7808	0.5796	0.6251	0.5773	0.9211	0.9203	0.8390	0.7940	0.7840	0.9211	0.8460	0.8460	0.8390	0.8531	0.9663	0.7257	
numeric_missing_10	numeric_missing	0.1000	0.1010	TEST	NTD2	6766	0.7923	0.8464	0.7804	0.7786	0.5817	0.6233	0.5737	0.9179	0.9166	0.8392	0.7923	0.7818	0.9179	0.8464	0.8464	0.8392	0.8536	0.9693	0.7234	
numeric_missing_10	numeric_missing	0.1000	0.1000	VAL	Wii21	304684	1.0000	1.0000	0.9999	1.0000	0.0001	0.9999	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
numeric_missing_10	numeric_missing	0.1000	0.1004	TEST	Wii21	305754	1.0000	1.0000	0.9999	1.0000	0.0001	0.9999	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
numeric_missing_10	numeric_missing	0.1000	0.1000	VAL	global_weighted	356002	0.9803	0.9784	0.9278	0.9416	0.0620	0.8953	0.8848	0.9921	0.9525	0.9816	0.9803	0.9780	0.9921	0.9921	0.9784	0.9784	0.9816	0.9752	0.9776	0.9792
numeric_missing_10	numeric_missing	0.1000	0.1002	TEST	global_weighted	356738	0.9800	0.9787	0.9259	0.9400	0.0636	0.8933	0.8818	0.9917	0.9511	0.9814	0.9800	0.9775	0.9917	0.9787	0.9787	0.9814	0.9760	0.9784	0.9789	
numeric_missing_20	numeric_missing	0.2000	0.0000	VAL	I23Sub	0	0.9185	0.9592	0.5087	0.4958	0.1088	0.1263	0.0314	0.9955	0.9831	0.9251	0.9185	0.8809	0.9955	0.9955	0.9592	0.9592	0.9251	0.9932	1.0000	0.9184
numeric_missing_20	numeric_missing	0.2000	0.0000	TEST	I23Sub	0	0.9188	0.9593	0.5121	0.5024	0.1111	0.1492	0.0435	0.9930	0.9730	0.9254	0.9188	0.8818	0.9930	0.9930	0.9593	0.9593	0.9254	0.9932	1.0000	0.9186
numeric_missing_20	numeric_missing	0.2000	0.1966	VAL	K99Sub	12010	0.9166	0.9213	0.9133	0.9156	0.3354	0.8346	0.8315	0.9726	0.9720	0.9192	0.9166	0.9162	0.9726	0.9726	0.9213	0.9213	0.9192	0.9234	0.8908	0.9519
numeric_missing_20	numeric_missing	0.2000	0.2015	TEST	K99Sub	12183	0.9166	0.9196	0.9140	0.9158	0.3405	0.8336	0.8318	0.9718	0.9709	0.9181	0.9166	0.9164	0.9718	0.9718	0.9196	0.9196	0.9181	0.9211	0.8973	0.9420
numeric_missing_20	numeric_missing	0.2000	0.1985	VAL	NTD1	76714	0.9416	0.9299	0.6703	0.7345	0.1812	0.5412	0.4756	0.9748	0.8035	0.9403	0.9416	0.9291	0.9748	0.9299	0.9299	0.9403	0.9196	0.9175	0.9424	
numeric_missing_20	numeric_missing	0.2000	0.2002	TEST	NTD1	77346	0.9392	0.9277	0.6555	0.7175	0.1879	0.5157	0.4431	0.9736	0.7929	0.9378	0.9392	0.9277	0.9736	0.9277	0.9277	0.9378	0.9175	0.9154	0.9399	
numeric_missing_20	numeric_missing	0.2000	0.1978	VAL	NTD2	13248	0.7940	0.8460	0.7823	0.7808	0.5796	0.6251	0.5773	0.9227	0.9217	0.8390	0.7940	0.7940	0.9227	0.8460	0.8460	0.8390	0.8531	0.9663	0.7257	
numeric_missing_20	numeric_missing	0.2000	0.1993	TEST	NTD2	13354	0.7923	0.8464	0.7804	0.7786	0.5817	0.6233	0.5737	0.9200	0.9185	0.8392	0.7923	0.7818	0.9200	0.8464	0.8464	0.8392	0.8536	0.9693	0.7234	
numeric_missing_20	numeric_missing	0.2000	0.2000	VAL	Wii21	609108	1.0000	1.0000	0.9999	1.0000	0.0001	0.9999	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
numeric_missing_20	numeric_missing	0.2000	0.2000	TEST	Wii21	609305	1.0000	1.0000	0.9999	1.0000	0.0001	0.9999	0.9999	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
numeric_missing_20	numeric_missing	0.2000	0.1998	VAL	global_weighted	711080	0.9778	0.9776	0.9117	0.9257	0.0681	0.8714	0.8540	0.9914	0.9533	0.9791	0.9776	0.9776	0.9914	0.9776	0.9776	0.9914	0.9761	0.9790	0.9762	
numeric_missing_20	numeric_missing	0.2000	0.2001	TEST	global_weighted	712188	0.9772	0.9771	0.9084	0.9220	0.0677	0.8660	0.8468	0.9910	0.9507	0.9786	0.9772	0.9733	0.9910	0.9771	0.9771	0.9786	0.9756	0.9787	0.9755	
categorical_oob_10	categorical_oob	0.1000	0.0973	VAL	I23Sub	8109	0.9234	0.9614	0.5382	0.5509	0.1242	0.2655	0.1317	0.9947	0.9837	0.9293	0.9234	0.8921	0.9947	0.9947	0.9614	0.9614	0.9293	0.9936	1.0000	0.9229
categorical_oob_10	categorical_oob	0.1000	0.0990	TEST	I23Sub	8250	0.9240	0.9617	0.5433	0.5597	0.1256	0.2826	0.1479	0.9923	0.9761	0.9298	0.9240	0.8935	0.9923	0.9923	0.9617	0.9617	0.9298	0.9936	1.0000	0.9234
categorical_oob_10	categorical_oob	0.1000	0.1008	VAL	K99Sub	2282	0.9159	0.9264	0.9109	0.9143	0.2920	0.8372	0.8294	0.9812	0.9796	0.9229	0.9159	0.9151	0.9812	0.9264	0.9264	0.9229	0.9300	0.8741	0.9788	
categorical_oob_10	categorical_oob	0.1000	0.0963	TEST	K99Sub	2183	0.9196	0.9312	0.9144	0.9180	0.2935	0.8455	0.8368	0.9801	0.9773	0.9274	0.9196	0.9801	0.9312	0.9312	0.9274	0.9350	0.8752	0.9873		
categorical_oob_10	categorical_oob	0.1000	0.1000	VAL	NTD1	22074	0.9574	0.9386	0.7720	0.8319	0.1581	0.6908	0.6654	0.9771	0.8028	0.9560	0.9574	0.9525	0.9771	0.9771	0.9386	0.9386	0.9560	0.9212	0.9176	0.9596
categorical_oob_10	categorical_oob	0.1000	0.1008	TEST	NTD1	22213	0.9550	0.9367	0.7565	0.8186	0.1659	0.6694	0.6393	0.9759	0.7871	0.9535	0.9550	0.9491	0.9759	0.9759	0.9367	0.9367	0.9535	0.9200	0.9166	0.9569
categorical_oob_10	categorical_oob	0.1000	0.1007	VAL	NTD2	21076	0.7374	0.8168	0.7221	0.7107	0.6040	0.5305	0.4574	0.9008	0.9024	0.8083	0.7374	0.7159	0.9008	0.9008	0.8168	0.8168	0.8083	0.8252	0.9614	0.6722
categorical_oob_10	categorical_oob	0.1000	0.0998	TEST	NTD2	20883	0.7350	0.8193	0.7193	0.7068	0.6055	0.5293	0.4520	0.9000	0.8980	0.8106	0.7350	0.7121	0.9000	0.9000	0.8193	0.8193	0.8106	0.8281	0.9695	0.6692
categorical_oob_10	categorical_oob	0.1000	0.0997	VAL	Wii21	214375	0.9996	0.9985	0.9996	0.9985	0.0020	0.9971	0.9971	1.0000	1.0000	0.9996	0.9996	0.9996	1.0000	0.9996	0.9975	0.9975	0.9996	0.9954	1.0000	0.9950
categorical_oob_10	categorical_oob	0.1000	0.0998	TEST	Wii21	214480	0.9996	0.9974	0.9996	0.9985	0.0021	0.9969	0.9969	1.0000	1.0000	0.9996	0.9996	0.9996	1.0000	0.9974	0.9974	0.9996	0.9951	1.0000	0.9947	
categorical_oob_10	categorical_oob	0.1000	0.0997	VAL	global_weighted	267916	0.9792	0.9769	0.9323	0.9446	0.0627	0.9012	0.8912	0.9913	0.9526	0.9814	0.9792	0.9769	0.9913	0.9769	0.9769	0.9814	0.9723	0.9786	0.9751	
categorical_oob_10	categorical_oob	0.1000	0.0998	TEST	global_weighted	268009	0.9786	0.9765	0.9289	0.9417	0.0646	0.8967	0.8855	0.9909	0.9489	0.9810	0.9786	0.9761	0.9909	0.9909	0.9765	0.9765	0.9810	0.9720	0.9786	0.9744
feature_dropout_10	feature_dropout	0.1000	0.1003	VAL	I23Sub	8361	0.9254	0.9624	0.5503	0.5719	0.1237	0.3052	0.1704	0.9954	0.9837	0.9310	0.9254	0.8964	0.9954	0.9954	0.9624	0.9624	0.9310	0.9938	1.0000	0.9248
feature_dropout_10	feature_dropout	0.1000	0.1013	TEST	I23Sub	8440	0.9248	0.9621	0.5484	0.5686	0.1249	0.2992	0.1644	0.9922	0.9753	0.9305	0.9248	0.8954	0.9922	0.9621	0.9621	0.9305	0.9937	1.0000	0.9242	
feature_dropout_10	feature_dropout	0.1000	0.1019	VAL	K99Sub	8473	0.9143	0.9215	0.9101	0.9130	0.3187	0.8315	0.8264	0.9762	0.9744	0.9187	0.9143	0.9137	0.9762	0.9762	0.9215	0.9215	0.9187	0.9242	0.8808	0.9621
feature_dropout_10	feature_dropout	0.1000	0.0991	TEST	K99Sub	8243	0.9177	0.9237	0.9140	0.9166	0.3186	0.8377	0.8335	0.9758	0.9750	0.9213	0.9177	0.9172	0.9758	0.9758	0.9237	0.9237	0.9213	0.9262	0.8874	0.9601
feature_dropout_10	feature_dropout	0.1000	0.0997	VAL	NTD1	60524	0.9453	0.9309	0.6945	0.7601	0.1716	0.5790	0.5251	0.9741	0.8031	0.9438	0.9453	0.9350	0.9741	0.9741	0.9309	0.9309	0.9438	0.9181	0.9155	0.9464
feature_dropout_10	feature_dropout	0.1000	0.0999	TEST	NTD1	60636	0.9437	0.9301	0.6843	0.7495	0.1780	0.5631	0.5045	0.9738	0.8003	0.9422	0.9437	0.9325	0.9738	0.9738	0.9301	0.9301	0.9422	0.9180	0.9155	0.9447
feature_dropout_10	feature_dropout	0.1000	0.1002	VAL	NTD2	27698	0.7348	0.8172	0.7192	0.7069	0.6043	0.5274	0.4516	0.9045	0.9049	0.8085	0.73									

18-training-label-noise-robustness-validation

FINAL REPORT - TRAINING LABEL-NOISE ROBUSTNESS VALIDATION

variant	target_label_noise_rate	actual_label_noise_rate	split	dataset	noisy_train_labels	acc	prec_macro	rec_macro	f1_macro	logloss	mcc	kappa	auc_roc_macro	pr_auc	prec_weighted	rec_weighted	f1_weighted	auc_roc_micro	auc_roc_weighted	ppv_macro	npv_macro	ppv_weighted	npv_weighted	ppv_positive	npv_negative
label_noise_05	0.0500	0.0500	VAL	I23Sub	810	0.9378	0.9682	0.6250	0.6836	0.3104	0.4839	0.3794	0.9973	0.9875	0.9417	0.9378	0.9201	0.9973	0.9973	0.9682	0.9682	0.9417	0.9947	1.0000	0.9365
label_noise_05	0.0500	0.0500	TEST	I23Sub	810	0.9352	0.9670	0.6107	0.6642	0.3088	0.4548	0.3428	0.9956	0.9784	0.9395	0.9352	0.9156	0.9956	0.9956	0.9670	0.9670	0.9395	0.9945	1.0000	0.9340
label_noise_05	0.0500	0.0500	VAL	K99Sub	882	0.9177	0.9191	0.9207	0.9177	0.5674	0.8398	0.8357	0.9742	0.9707	0.9223	0.9177	0.9178	0.9742	0.9742	0.9191	0.9191	0.9223	0.9159	0.9656	0.8726
label_noise_05	0.0500	0.0500	TEST	K99Sub	882	0.9116	0.9135	0.9148	0.9116	0.5667	0.8283	0.8237	0.9744	0.9705	0.9169	0.9166	0.9117	0.9744	0.9744	0.9135	0.9135	0.9169	0.9102	0.9631	0.8639
label_noise_05	0.0500	0.0500	VAL	NTD1	12881	0.9665	0.9409	0.8328	0.8776	0.1195	0.7660	0.7557	0.9887	0.8965	0.9652	0.9665	0.9642	0.9887	0.9887	0.9409	0.9409	0.9652	0.9165	0.9115	0.9702
label_noise_05	0.0500	0.0500	TEST	NTD1	12881	0.9636	0.9365	0.8161	0.8646	0.1234	0.7429	0.7299	0.9886	0.8907	0.9621	0.9636	0.9607	0.9886	0.9886	0.9365	0.9365	0.9621	0.9109	0.9057	0.9673
label_noise_05	0.0500	0.0500	VAL	NTD2	1954	0.8309	0.8639	0.8219	0.8238	0.6124	0.6846	0.6549	0.9323	0.9226	0.8583	0.8309	0.8258	0.9323	0.9323	0.8639	0.8639	0.8583	0.8696	0.9599	0.7679
label_noise_05	0.0500	0.0500	TEST	NTD2	1954	0.8248	0.8628	0.8152	0.8165	0.6130	0.6763	0.6420	0.9334	0.9275	0.8567	0.8248	0.8188	0.9334	0.9334	0.8628	0.8628	0.8567	0.8689	0.9665	0.7591
label_noise_05	0.0500	0.0500	VAL	WII21	41806	0.9995	0.9996	0.9969	0.9982	0.0242	0.9965	0.9965	1.0000	1.0000	0.9995	0.9995	0.9995	1.0000	1.0000	0.9996	0.9996	0.9995	0.9996	0.9995	0.9996
label_noise_05	0.0500	0.0500	TEST	WII21	41806	0.9996	0.9997	0.9987	0.9977	0.0241	0.9973	0.9973	1.0000	1.0000	0.9996	0.9996	0.9996	1.0000	1.0000	0.9997	0.9997	0.9996	0.9997	0.9996	0.9997
label_noise_05	0.0500	0.0500	VAL	global_weighted	58333	0.9845	0.9804	0.9485	0.9602	0.0772	0.9257	0.9209	0.9948	0.9739	0.9853	0.9845	0.9836	0.9948	0.9948	0.9804	0.9804	0.9853	0.9756	0.9782	0.9826
label_noise_05	0.0500	0.0500	TEST	global_weighted	58333	0.9836	0.9794	0.9448	0.9570	0.0779	0.9203	0.9146	0.9948	0.9727	0.9845	0.9836	0.9825	0.9948	0.9948	0.9794	0.9794	0.9845	0.9743	0.9772	0.9815
label_noise_10	0.1000	0.1000	VAL	I23Sub	1620	0.9171	0.4585	0.5000	0.4784	0.3874	0.0000	0.0000	0.9290	0.7707	0.8410	0.9171	0.8774	0.9290	0.9290	0.4585	0.4585	0.8410	0.0000	0.9171	
label_noise_10	0.1000	0.1000	TEST	I23Sub	1620	0.9168	0.4584	0.5000	0.4783	0.3865	0.0000	0.0000	0.9167	0.7208	0.8405	0.9168	0.8770	0.9167	0.9167	0.4584	0.4584	0.8405	0.0763	0.0000	0.9168
label_noise_10	0.1000	0.1000	VAL	K99Sub	1763	0.8566	0.8935	0.8462	0.8499	0.4958	0.7382	0.7081	0.9808	0.9715	0.8864	0.8566	0.8521	0.9808	0.9808	0.8935	0.8935	0.8864	0.9007	0.7887	0.9984
label_noise_10	0.1000	0.1000	TEST	K99Sub	1763	0.8611	0.8967	0.8510	0.8549	0.4955	0.7464	0.7155	0.9791	0.9711	0.8898	0.8611	0.8569	0.9791	0.9791	0.8967	0.8967	0.8898	0.9037	0.7935	1.0000
label_noise_10	0.1000	0.1000	VAL	NTD1	25761	0.9150	0.4575	0.5000	0.4778	0.1873	0.0000	0.0000	0.9442	0.8559	0.8372	0.9150	0.8744	0.9442	0.9442	0.4575	0.4575	0.8372	0.0778	0.0000	0.9150
label_noise_10	0.1000	0.1000	TEST	NTD1	25761	0.9150	0.4575	0.5000	0.4778	0.1934	0.0000	0.0000	0.9430	0.8461	0.8372	0.9150	0.8744	0.9430	0.9430	0.4575	0.4575	0.8372	0.0778	0.0000	0.9150
label_noise_10	0.1000	0.1000	VAL	NTD2	3908	0.8633	0.8721	0.8587	0.8610	0.6066	0.7307	0.7233	0.9660	0.9499	0.8695	0.8633	0.8620	0.9660	0.9660	0.8721	0.8721	0.8695	0.8748	0.9168	0.8275
label_noise_10	0.1000	0.1000	TEST	NTD2	3908	0.8639	0.8744	0.8589	0.8613	0.6076	0.7332	0.7243	0.9696	0.9588	0.8715	0.8639	0.8624	0.9696	0.9696	0.8744	0.8744	0.8715	0.8773	0.9243	0.8245
label_noise_10	0.1000	0.1000	VAL	WII21	83612	0.9994	0.9996	0.9963	0.9979	0.0716	0.9959	0.9959	1.0000	1.0000	0.9994	0.9994	0.9994	1.0000	1.0000	0.9996	0.9996	0.9994	0.9997	0.9994	0.9998
label_noise_10	0.1000	0.1000	TEST	WII21	83612	0.9993	0.9995	0.9955	0.9975	0.0719	0.9950	0.9950	1.0000	1.0000	0.9993	0.9993	0.9993	1.0000	1.0000	0.9995	0.9995	0.9993	0.9997	0.9993	0.9998
label_noise_10	0.1000	0.1000	VAL	global_weighted	116664	0.9729	0.8665	0.8729	0.8691	0.1259	0.7494	0.7486	0.9853	0.9629	0.9554	0.9729	0.9633	0.9853	0.9853	0.8665	0.8665	0.9554	0.7776	0.7589	0.9741
label_noise_10	0.1000	0.1000	TEST	global_weighted	116664	0.9729	0.8666	0.8725	0.8688	0.1275	0.7490	0.7482	0.9849	0.9603	0.9554	0.9729	0.9633	0.9849	0.9849	0.8666	0.8666	0.9554	0.7778	0.7591	0.9740
label_noise_30	0.3000	0.3000	VAL	I23Sub	4860	0.9816	0.9901	0.8889	0.9325	0.4223	0.8732	0.8652	0.9764	0.9396	0.9819	0.9816	0.9805	0.9764	0.9764	0.9901	0.9901	0.9819	1.0000	0.9803	
label_noise_30	0.3000	0.3000	TEST	I23Sub	4860	0.9798	0.9892	0.8789	0.9257	0.4202	0.8611	0.8516	0.9799	0.9437	0.9603	0.9798	0.9786	0.9799	0.9799	0.9892	0.9892	0.9803	0.9982	1.0000	0.9785
label_noise_30	0.3000	0.3000	VAL	K99Sub	5290	0.8830	0.9018	0.8758	0.8797	0.5689	0.7773	0.7618	0.9559	0.9356	0.8970	0.8830	0.8811	0.9559	0.9559	0.9018	0.9018	0.8970	0.9067	0.8303	0.9734
label_noise_30	0.3000	0.3000	TEST	K99Sub	5290	0.8809	0.8990	0.8739	0.8777	0.5702	0.7724	0.7576	0.9511	0.9284	0.8943	0.8809	0.8790	0.9511	0.9511	0.8990	0.8990	0.8943	0.9037	0.8293	0.9686
label_noise_30	0.3000	0.3000	VAL	NTD1	77283	0.9150	0.4575	0.5000	0.4778	0.2734	0.0000	0.0000	0.9811	0.8720	0.8372	0.9150	0.8744	0.9811	0.9811	0.4575	0.4575	0.8372	0.0778	0.0000	0.9150
label_noise_30	0.3000	0.3000	TEST	NTD1	77283	0.9150	0.4575	0.5000	0.4778	0.2755	0.0000	0.0000	0.9790	0.8578	0.8372	0.9150	0.8744	0.9790	0.9790	0.4575	0.4575	0.8372	0.0778	0.0000	0.9150
label_noise_30	0.3000	0.3000	VAL	NTD2	11723	0.7854	0.8143	0.7760	0.7759	0.6275	0.5891	0.5617	0.9375	0.9170	0.8096	0.7854	0.7786	0.9375	0.9375	0.8143	0.8143	0.8096	0.8191	0.8962	0.7325
label_noise_30	0.3000	0.3000	TEST	NTD2	11723	0.7932	0.8238	0.7837	0.7839	0.6270	0.6062	0.5775	0.9422	0.9234	0.8188	0.7932	0.7865	0.9422	0.9422	0.8238	0.8238	0.8188	0.8288	0.9099	0.7378
label_noise_30	0.3000	0.3000	VAL	WII21	250837	0.9271	0.4636	0.5000	0.4811	0.2651	0.0000	0.0000	0.9982	0.9999	0.8596	0.9982	0.8921	0.9982	0.9982	0.4636	0.4636	0.8596	0.0675	0.9271	0.0000
label_noise_30	0.3000	0.3000	TEST	WII21	250837	0.9272	0.4636	0.5000	0.4811	0.2652	0.0000	0.0000	0.9981	0.9998	0.8596	0.9981	0.8921	0.9981	0.9981	0.4636	0.4636	0.8596	0.0675	0.9272	0.0000
label_noise_30	0.3000	0.3000	VAL	global_weighted	349993	0.9198	0.4879	0.5203	0.5025	0.2859	0.0436	0.0423	0.9915	0.9670	0.8553	0.9198	0.8854	0.9915	0.9915	0.4879	0.4879	0.8553	0.1206	0.7209	0.2549
label_noise_30	0.3000	0.3000	TEST	global_weighted	349993	0.9200	0.4882	0.5204	0.5027	0.2864	0.0439	0.0426	0.9910	0.9641	0.8555	0.9200	0.8857	0.9910	0.9910	0.4882	0.4882	0.8555	0.1209	0.7214	0.2550

19-shared-categorical-hash-bucket-sensitivity

FINAL REPORT - SHARED CATEGORICAL HASH-BUCKET SENSITIVITY VALIDATION

variant	hash_buckets	split	dataset	shared_categorical_features	acc	prec_macro	rec_macro	f1_macro	logloss	mcc	kappa	auc_roc_macro	pr_auc	prec_weighted	rec_weighted	f1_weighted	auc_roc_micro	auc_roc_weighted	ppv_macro	npv_macro	ppv_weighted	npv_weighted	ppv_positive	npv_negative
hash_1024	1024	VAL	I23Sub		0	0.9905	0.9949	0.9427	0.9670	0.1059	0.9361	0.9341	0.9935	0.9703	0.9906	0.9905	0.9902	0.9935	0.9949	0.9949	0.9906	0.9991	1.0000	0.9897
hash_1024	1024	TEST	I23Sub		0	0.9879	0.9935	0.9273	0.9575	0.1080	0.9184	0.9151	0.9911	0.9609	0.9881	0.9879	0.9875	0.9911	0.9935	0.9935	0.9881	0.9989	1.0000	0.9870
hash_1024	1024	VAL	K99Sub		1	0.9103	0.9196	0.9056	0.9087	0.2749	0.8250	0.8182	0.9787	0.9780	0.9163	0.9103	0.9095	0.9787	0.9787	0.9196	0.9163	0.9228	0.8720	0.9671
hash_1024	1024	TEST	K99Sub		1	0.9106	0.9200	0.9058	0.9090	0.2737	0.8257	0.8187	0.9800	0.9801	0.9168	0.9106	0.9098	0.9800	0.9800	0.9200	0.9168	0.9233	0.8716	0.9684
hash_1024	1024	VAL	NTD1		2	0.9150	0.4575	0.5000	0.4778	0.1647	0.0000	0.0000	0.9832	0.8540	0.8372	0.9150	0.8744	0.9832	0.4575	0.4575	0.8372	0.0778	0.0000	0.9150
hash_1024	1024	TEST	NTD1		2	0.9150	0.4575	0.5000	0.4778	0.1728	0.0000	0.0000	0.9819	0.8426	0.8372	0.9150	0.8744	0.9819	0.4575	0.4575	0.8372	0.0778	0.0000	0.9150
hash_1024	1024	VAL	NTD2		0	0.8600	0.8671	0.8559	0.8580	0.4989	0.7229	0.7170	0.9236	0.9169	0.8649	0.8600	0.8590	0.9236	0.8671	0.8671	0.8649	0.8693	0.9054	0.8288
hash_1024	1024	TEST	NTD2		0	0.8608	0.8703	0.8560	0.8583	0.5003	0.7262	0.7181	0.9220	0.9131	0.8676	0.8608	0.8594	0.9220	0.8703	0.8703	0.8676	0.8731	0.9171	0.8236
hash_1024	1024	VAL	WII21		0	1.0000	1.0000	1.0000	1.0000	0.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
hash_1024	1024	TEST	WII21		0	1.0000	1.0000	1.0000	1.0000	0.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
hash_1024	1024	VAL	global_weighted		3	0.9751	0.8745	0.8825	0.8781	0.0587	0.7664	0.7660	0.9933	0.9642	0.9581	0.9751	0.9660	0.9933	0.8745	0.8745	0.9581	0.7908	0.7741	0.9749
hash_1024	1024	TEST	global_weighted		3	0.9750	0.8746	0.8823	0.8780	0.0606	0.7662	0.7658	0.9930	0.9615	0.9582	0.9750	0.9660	0.9930	0.8746	0.8746	0.9582	0.7909	0.7745	0.9747
hash_8192	8192	VAL	I23Sub		1	0.9931	0.9963	0.9583	0.9764	0.2225	0.9538	0.9528	0.9816	0.9797	0.9931	0.9931	0.9930	0.9816	0.9963	0.9963	0.9931	0.9994	1.0000	0.9925
hash_8192	8192	TEST	I23Sub		1	0.9931	0.9963	0.9585	0.9765	0.2238	0.9540	0.9529	0.9693	0.9675	0.9931	0.9931	0.9930	0.9693	0.9963	0.9963	0.9931	0.9994	1.0000	0.9925
hash_8192	8192	VAL	K99Sub		1	0.9071	0.9075	0.9094	0.9070	0.3820	0.8169	0.8144	0.9729	0.9690	0.9101	0.9071	0.9072	0.9729	0.9075	0.9075	0.9101	0.9049	0.9460	0.8691
hash_8192	8192	TEST	K99Sub		1	0.9074	0.9083	0.9100	0.9073	0.3759	0.8183	0.8150	0.9768	0.9743	0.9112	0.9074	0.9075	0.9768	0.9083	0.9083	0.9112	0.9054	0.9508	0.8658
hash_8192	8192	VAL	NTD1		2	0.9669	0.9434	0.8329	0.8786	0.1324	0.7684	0.7577	0.9751	0.7999	0.9657	0.9669	0.9645	0.9751	0.9434	0.9434	0.9657	0.9211	0.9165	0.9702
hash_8192	8192	TEST	NTD1		2	0.9641	0.9402	0.8164	0.8660	0.1396	0.7464	0.7327	0.9736	0.7883	0.9627	0.9641	0.9612	0.9736	0.9402	0.9402	0.9627	0.9177	0.9131	0.9673
hash_8192	8192	VAL	NTD2		0	0.7951	0.8440	0.8060	0.7910	0.5177	0.6489	0.5981	0.9370	0.8937	0.8525	0.7951	0.7893	0.9370	0.8440	0.8440	0.8525	0.8355	0.6988	0.9892
hash_8192	8192	TEST	NTD2		0	0.7922	0.8420	0.8032	0.7880	0.5160	0.6441	0.5926	0.9442	0.9074	0.8506	0.7922	0.7862	0.9442	0.8420	0.8420	0.8506	0.8335	0.6960	0.9880
hash_8192	8192	VAL	WII21		0	1.0000	1.0000	1.0000	1.0000	0.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
hash_8192	8192	TEST	WII21		0	1.0000	1.0000	1.0000	1.0000	0.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
hash_8192	8192	VAL	global_weighted		4	0.9843	0.9808	0.9547	0.9645	0.0555	0.9337	0.9296	0.9860	0.9917	0.9515	0.9860	0.9843	0.9836	0.9917	0.9808	0.9860	0.9756	0.9707	0.9910
hash_8192	8192	TEST	global_weighted		4	0.9836	0.9801	0.9509	0.9616	0.0569	0.9287	0.9239	0.9915	0.9493	0.9853	0.9836	0.9828	0.9915	0.9801	0.9801	0.9853	0.9748	0.9699	0.9902

20-cross-feature-interaction-necessity

FINAL REPORT - CROSS-FEATURE INTERACTION NECESSITY VALIDATION

variant	split	dataset	cross_features_generated	acc	prec_macro	rec_macro	f1_macro	logloss	mcc	kappa	auc_roc_macro	pr_auc	prec_weighted	rec_weighted	f1_weighted	auc_roc_micro	auc_roc_weighted	ppv_macro	npv_macro	ppv_weighted	npv_weighted	ppv_positive	npv_negative
no_cross_features	VAL	I23Sub	0	0.9171	0.4585	0.5000	0.4784	0.1072	0.0000	0.0000	0.9984	0.9865	0.8410	0.9171	0.8774	0.9984	0.9984	0.4585	0.4585	0.8410	0.0761	0.0000	0.9171
no_cross_features	TEST	I23Sub	0	0.9171	0.9585	0.5017	0.4818	0.1101	0.0563	0.0063	0.9955	0.9878	0.9239	0.9171	0.8777	0.9955	0.9955	0.9585	0.9585	0.9239	0.9931	1.0000	0.9170
no_cross_features	VAL	K99Sub	0	0.8746	0.9039	0.8656	0.8698	0.3608	0.7685	0.7437	0.9717	0.9674	0.8975	0.8746	0.8715	0.9717	0.9717	0.9039	0.9039	0.8975	0.9102	0.8108	0.9969
no_cross_features	TEST	K99Sub	0	0.8748	0.9048	0.8658	0.8700	0.3611	0.7696	0.7442	0.9703	0.9666	0.8984	0.8748	0.8717	0.9703	0.9703	0.9048	0.9048	0.8984	0.9111	0.8103	0.9992
no_cross_features	VAL	NTD1	0	0.9659	0.9349	0.8335	0.8760	0.1814	0.7616	0.7524	0.9682	0.8128	0.9644	0.9659	0.9636	0.9682	0.9682	0.9349	0.9349	0.9644	0.9054	0.8993	0.9704
no_cross_features	TEST	NTD1	0	0.9631	0.9309	0.8172	0.8635	0.1943	0.7394	0.7276	0.9651	0.7972	0.9613	0.9631	0.9603	0.9651	0.9651	0.9309	0.9309	0.9613	0.9005	0.8942	0.9675
no_cross_features	VAL	NTD2	0	0.5291	0.2646	0.4999	0.3460	0.5879	-0.0103	-0.0002	0.9284	0.9074	0.2801	0.5291	0.3663	0.9284	0.9284	0.2646	0.2646	0.2801	0.2491	0.0000	0.5292
no_cross_features	TEST	NTD2	0	0.5290	0.2646	0.4998	0.3460	0.5881	-0.0146	-0.0005	0.9285	0.9086	0.2801	0.5290	0.3662	0.9285	0.9285	0.2646	0.2646	0.2801	0.2491	0.0000	0.5291
no_cross_features	VAL	WII21	0	1.0000	0.9998	1.0000	0.9999	0.0002	0.9998	0.9998	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
no_cross_features	TEST	WII21	0	1.0000	1.0000	0.9997	0.9998	0.0004	0.9997	0.9997	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
no_cross_features	VAL	global_weighted	0	0.9736	0.9520	0.9374	0.9414	0.0668	0.8960	0.8939	0.9901	0.9549	0.9642	0.9736	0.9671	0.9901	0.9901	0.9520	0.9520	0.9642	0.9398	0.9275	0.9765
no_cross_features	TEST	global_weighted	0	0.9730	0.9581	0.9337	0.9387	0.0699	0.8916	0.8884	0.9894	0.9515	0.9647	0.9730	0.9663	0.9894	0.9894	0.9581	0.9581	0.9647	0.9514	0.9402	0.9759
pairwise_cross_only	VAL	I23Sub	15	0.9522	0.9752	0.7118	0.7849	0.1469	0.6345	0.5741	0.9998	0.9977	0.9546	0.9522	0.9431	0.9998	0.9998	0.9752	0.9752	0.9546	0.9959	1.0000	0.9504
pairwise_cross_only	TEST	I23Sub	15	0.9531	0.9756	0.7180	0.7911	0.1434	0.6440	0.5863	0.9974	0.9905	0.9553	0.9531	0.9444	0.9974	0.9974	0.9756	0.9756	0.9553	0.9959	1.0000	0.9513
pairwise_cross_only	VAL	K99Sub	3	0.9378	0.9407	0.9355	0.9372	0.3712	0.8761	0.8745	0.9730	0.9681	0.9391	0.9378	0.9376	0.9730	0.9730	0.9407	0.9407	0.9391	0.9422	0.9183	0.9630
pairwise_cross_only	TEST	K99Sub	3	0.9357	0.9389	0.9332	0.9350	0.3748	0.8721	0.8702	0.9708	0.9637	0.9373	0.9357	0.9355	0.9708	0.9708	0.9389	0.9389	0.9373	0.9406	0.9145	0.9634
pairwise_cross_only	VAL	NTD1	3	0.9669	0.9434	0.8329	0.8786	0.1635	0.7684	0.7577	0.9675	0.7884	0.9657	0.9669	0.9645	0.9675	0.9675	0.9434	0.9434	0.9657	0.9211	0.9165	0.9702
pairwise_cross_only	TEST	NTD1	3	0.9641	0.9402	0.8164	0.8660	0.1718	0.7464	0.7327	0.9664	0.7756	0.9627	0.9641	0.9612	0.9664	0.9664	0.9402	0.9402	0.9627	0.9177	0.9131	0.9673
pairwise_cross_only	VAL	NTD2	15	0.8283	0.8600	0.8370	0.8265	0.6041	0.6967	0.6617	0.9700	0.9555	0.8672	0.8283	0.8255	0.9700	0.9700	0.8600	0.8600	0.8672	0.8529	0.7375	0.9826
pairwise_cross_only	TEST	NTD2	15	0.8200	0.8553	0.8293	0.8179	0.6064	0.6841	0.6458	0.9708	0.9554	0.8628	0.8200	0.8167	0.9708	0.9708	0.8553	0.8553	0.8628	0.8478	0.7278	0.9828
pairwise_cross_only	VAL	WII21	10	1.0000	1.0000	1.0000	1.0000	0.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
pairwise_cross_only	TEST	WII21	10	1.0000	1.0000	1.0000	1.0000	0.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
pairwise_cross_only	VAL	global_weighted	46	0.9853	0.9816	0.9527	0.9635	0.0640	0.9318	0.9274	0.9914	0.9513	0.9864	0.9853	0.9846	0.9914	0.9914	0.9816	0.9816	0.9864	0.9767	0.9715	0.9916
pairwise_cross_only	TEST	global_weighted	46	0.9844	0.9807	0.9488	0.9604	0.0659	0.9265	0.9214	0.9911	0.9483	0.9856	0.9844	0.9835	0.9911	0.9911	0.9807	0.9807	0.9856	0.9758	0.9704	0.9910